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Research Report

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Title of Research Report
Autoparametric Resonance in Elastic Pendulums

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Autoparametric Resonance in Elastic Pendulums

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Abstract

The harmonic oscillator is one of the most fundamental concepts in physics. Nevertheless, chaotic motion is easily observed even in very simple systems. Here, we explore the behaviour of the elastic pendulum, where a mass is suspended by a spring and set in motion.

The mathematical model, as formulated with an approximation of the Lagrangian, predicts two coupled oscillators. Experimental evidence points to resonant behaviour, where the amplitude of the pendulum oscillation grows if the natural frequency is in an integer ratio with the spring oscillation. However, the natural frequency of the pendulum will change as the amplitude of its oscillation increases. This is because the natural frequency of the pendulum is dependent on the length, and is changing due to the stretching of the spring. This forms a different type of periodic motion, where the amplitude of the horizontal oscillation grows and decays exponentially in turn. A very small initial deviation from the vertical axis results in an interchanging between horizontal and vertical oscillations.

In our experiment, we analysed the behaviour of the elastic pendulum by adjusting its length, and hence adjusting the ratio between the two angular frequencies. Through the development and analysis of the mathematical theory from first principles, we explain the origin for the resonant frequency responsible for the motion of the elastic pendulum. We can use this to make predictions with regards to the optimal length to cause resonant behaviour. This prediction is tested and confirmed via experimental data, a feature significantly lacking in existing literature.

Keywords: Resonance, Elastic Pendulum, Lagrangian Mechanics, Perturbation, Nonlinear Dynamics

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1 Introduction

1.1 Background

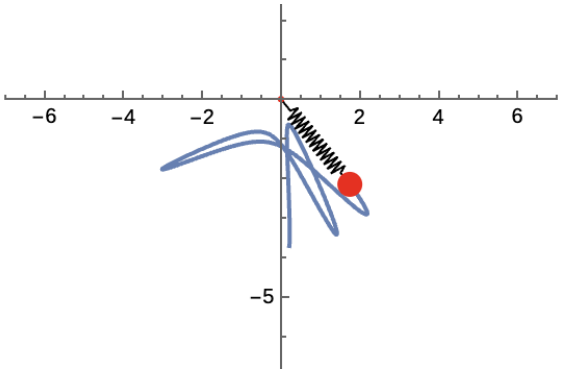


Figure 1: Simulated example of a mass, in red, attached via a spring. We can see its amplitude grow rapidly, from a very small initial horizontal displacement from the y-axis.[3]

The harmonic oscillator is arguably the most important singular mechanical system in physics, since many complex phenomena can be approximated as simple harmonic oscillators locally. However, very simple adjustments to the oscillator system can lead to very dramatic and sometimes chaotic motion.

In our experiment, we constructed a pendulum with an elastic rod by suspending a mass on a string fitted with a spring. When initially displaced, it spontaneously transitions between radial spring oscillations and tangential pendulum oscillations. This is rather surprising behaviour, as given the setup, one might simply expect the mass to continue bobbing up and down, as there is no obvious source of horizontal movement. However, the mass almost spontaneously goes from bobbing to swinging in a striking way, but only sometimes - it only happens at certain lengths.

This is an example of autoparametric resonance, where the oscillation of the spring system can excite the pendulum system when their periods are in resonance, specifically $\frac{\omega_p}{\omega_s} = \frac{1}{2}$, where the driving spring frequency is twice the natural pendulum frequency, and equivalently the spring is extended every time the pendulum returns to its equilibrium position [5]. The term parametric refers to the idea that this is a resonance between the parameters - in our case, the length of the pendulum and the angular velocity of the spring. Its prefix auto refers to the idea that this resonance is self-induced and not via an external driving force - in our case, the spring and pendulum system drive each other in an interchanging manner. This resonant motion can be described and understood mathematically, and it serves as the basis for the understanding of other analogous systems, in particular the behaviour of the CO_2 molecule [8]. An example of application includes the optical parametric oscillator, which breaks down a laser wave into two component parts of lower energy. This is a method used to generate entangled pairs of photons [6]. Parametric amplifiers are electronic components that also utilise the phenomenon of parametric resonance to amplify signals.

In particular, the motion of the elastic pendulum is interesting due to its chaotic nature. As seen in Figure 1, all horizontal displacements, however small, are amplified, leading to chaotic

motion. In practice, we can observe the mass consistently switching between a 'pendulum mode' and 'spring mode', that is its maximum horizontal and vertical motion respectively. In three dimensions, we can also observe a precession, as the plane of oscillation rotates by some amount each time it switches from spring mode to pendulum mode.

1.2 Overview

The switches between the two modes are very curious, and have some unique properties that can be modelled. One such property is the 'optimal' length, where energy is fully transferred to the pendulum/spring modes, and can be measured by the amplitude. This is equivalently a measure of the amount of energy transferred to the pendulum mode of oscillation. In this paper, the aim is to derive fully and experimentally confirm the optimal length that allows the maximum amplitude to be achieved. This is given by the 1:2 frequency ratio. Intuitively, this can be thought of as the spring pumping each time the mass swings, like two people pushing a child on a swing from each end. Then, we will apply this theory to our equipment, using the Tracker application to analyse the position of our oscillator for different lengths and confirm the theoretical prediction.

2 Theory

There are several works that have explored this topic in the past. However, given its complexity, it is often broken down into individual components, such as the setup of the Lagrangian, or the analysis of Mathieu's equation or precession. Additionally, these sources use very different variables, or can often be confusing by overt assumptions or skipped steps. Therefore, we present a complete, detailed derivation of the optimal 2:1 frequency ratio. This will be done by studying the Lagrangian of the system, which has the benefit of leading us to Mathieu's equation and its relevant theory.

2.1 Derivation

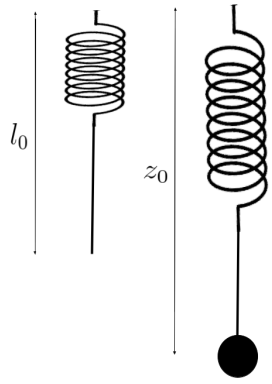


Figure 2: String attached to a spring with and without mass

Consider our system shown in Figure 2, where the combined length of the string and relaxed spring, l_0, is extended by a point mass m in a gravitational field, such that its new length is

$$z_0 = l_0 + \frac{mg}{k},$$

for a spring constant k. We can analyse the kinetic and potential energies to work towards an approximate Lagrangian for the 2D system. Damping is considered to be negligible, for a few reasons. Firstly, and most importantly, the motion happens in a short span of time. At a length where resonance occurs, the amplitude begins to grow immediately. Secondly, the density of the mass is quite large, and air drag is not a significant factor within the short timeframe. Finally, since we are comparing the same setup at different lengths, the amount of damping is essentially constant and thus will not affect experimental conclusions significantly. We define x and z to be the horizontal and vertical displacements (with positive being upwards), zeroed at the resting position of the mass. The kinetic energy, T, is then comprised of both the horizontal and vertical velocities:

$$T = \frac{m(\dot{x}^2 + \dot{z}^2)}{2}.$$

The potential energy, V, is comprised of a gravitational component, as well as elastic energy stored in the spring as derived from Hooke's law:

$$V = mgz + \frac{k(l_0 - l)^2}{2},$$

where l is the instantaneous length of the system (the sum length of the string and stretched spring). By applying the Pythagorean theorem in Figure 3, we also have

$$l^2 = (z_0 - z)^2 + x^2.$$

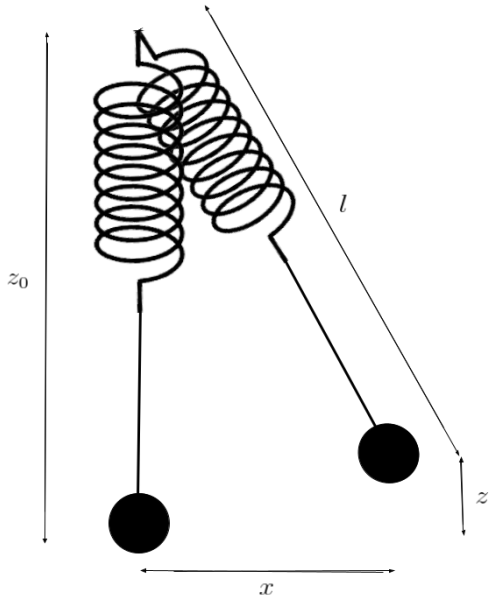


Figure 3: Pendulum in motion, where l, z, x are instantaneous.

We can also use the standard definitions for angular velocities in spring and pendulum systems, which are, respectively,

$$\omega_s^2 = \frac{k}{m} \qquad\qquad \omega_p^2 = \frac{g}{z_0}.$$

We can approximate by ignoring high orders of x and z. This is a subtle step, where we can substitute our expression for l into the expression for V, and expand the Taylor series of $\sqrt{1+x}$, ignoring terms above cubic factors of x and z as follows:

$$\begin{aligned}
V =& mgz + \frac{k(l_0 - l)^2}{2} \\
=& mgz + \frac{k}{2}l^2 - kl_0 l + \frac{k}{2}l_0^2 \\
=& mgz + \frac{k}{2}l_0^2 + \frac{1}{2}k(z_0^2 - 2z_0 z + z^2 + x^2) - kl_0\sqrt{z_0^2 - 2z_0 z + z^2 + x^2} \\
=& mgz + \frac{k}{2}l_0^2 + \frac{1}{2}k(z_0^2 - 2z_0 z + z^2 + x^2) - kl_0 z_0\sqrt{1 + (\frac{z^2+x^2}{z_0^2} - \frac{2z}{z_0})} \\
\approx& mgz + \frac{k}{2}l_0^2 + \frac{1}{2}k(z_0^2 - 2z_0 z + z^2 + x^2) - kl_0 z_0(1 + \frac{z^2+x^2}{2z_0^2} - \frac{z}{z_0} - \frac{1}{8}(\frac{z^2+x^2}{z_0^2} - \frac{2z}{z_0})^2) \\
&+ O((\frac{z^2+x^2}{z_0^2} - \frac{2z}{z_0})^3).
\end{aligned}$$

Further simplifying by grouping and cancelling terms of constants, z, z^2, x, and zx^2, then substituting w_p, w_s, and defining $\lambda = \frac{kl_0}{mz_0^2}$ for simplicity, we get

$$\begin{aligned}
V &\approx \frac{1}{2k}m^2g^2 + \frac{mg}{2z_0}x^2 + \frac{1}{2}kz^2 - \frac{k}{2}\frac{l_0}{z_0^2}zx^2 \\
&\approx m[\frac{mg^2}{2k} + \frac{w_p^2x^2}{2} + \frac{w_s^2z^2}{2} - \frac{kl_0}{mz_0^2}zx^2] \\
&\approx m[\frac{mg^2}{2k} + \frac{w_p^2x^2}{2} + \frac{w_s^2z^2}{2} - \frac{\lambda zx^2}{2}].
\end{aligned}$$

Then, we can arrive at the approximate Lagrangian:

$$\begin{aligned}
L = T - V &= \frac{m(\dot{x}^2 + \dot{z}^2)}{2} - [m(\frac{mg^2}{2k} + \frac{w_p^2x^2}{2} + \frac{w_s^2z^2}{2} - \frac{\lambda zx^2)}{2}] \\
&= m(\frac{\dot{x}^2 - \omega_p^2x^2}{2} + \frac{\dot{z}^2 - \omega_s^2z^2}{2} + \frac{\lambda zx^2}{2} + \frac{mg}{2k}).
\end{aligned}$$

The Lagrangian is characterised by simple harmonic motion in the x and z directions, as well as the final term including both variables, modeling the transfer of energy from horizontal pendulum to vertical spring oscillation. We can thus derive equations of motion from the Lagrangian via the Euler-Lagrange equation:

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x} \qquad\qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{z}} = \frac{\partial L}{\partial z}$$

$$\ddot{x} = \lambda xz - \omega_p^2 x \quad (1) \qquad\qquad \ddot{z} = \frac{\lambda x^2}{2} - \omega_s^2 z. \quad (2)$$

We can model the elastic pendulum as two different harmonic oscillators with a coupling term.

Consider the oscillation in the z-direction, or equivalently the spring as a normal simple harmonic

oscillator at small amplitudes ($x \ll z$). We can do this because the coupling term in (2) is negligible. We can then rewrite (2), which governs its radial motion. This approximate vertical motion model is

$$z = a\cos(\omega_s t).$$

We can then plug this into (1), yielding:

$$\ddot{x} + (\omega_p^2 - \frac{\omega_s^2 l_0}{z_0^2} a\cos(\omega_s t))x = 0. \tag{3}$$

This differential equation is known as Mathieu's equation, with its generalised form being

$$\ddot{x} + (\delta + \epsilon\cos t)x = 0. \tag{4}$$

This is a well-studied equation, with applications in many different fields of physics. We can solve this equation with a perturbation approach. Let us define a variable τ, where $\tau = \omega_s t$. Because

$$\frac{dx}{dt} = \frac{dx}{d\tau}\frac{d\tau}{dt} = \omega_s \frac{dx}{d\tau},$$

then by application of the chain rule we can see that

$$\frac{d^2x}{d\tau^2} = \omega_s^2 \frac{d^2x}{dt^2}.$$

We can thus factor out ω_s^2 from (3) to see

$$\ddot{x} + (\frac{w_p^2}{w_s^2} - \frac{l_0}{z_0^2} a\cos(\tau))x = 0. \tag{5}$$

Comparing this with (4), we can interpret δ as the square of the ratio between the two angular velocities, $\frac{\omega_p^2}{\omega_s^2}$. ϵ is assumed to be small, as $a \ll l_0$ and $l_0 \approx z_0$. It is important to note that this new variable τ is in the timescale of the spring oscillation. We can then let $\eta = \epsilon t$ where ϵ is some small parameter. Effectively, η is time slowed down; this will be useful for a separation of the fast dynamics in τ, in our case the individual oscillations of the oscillators, and the slow dynamics in η, the change in amplitude of the pendulum. This is known as multiple scales analysis, and through this rescaling of time we hope to derive certain conditions for our system such that the solutions are not unbounded. Here is where the perturbation happens - we consider x as a function of τ and ϵt, and then from (4):

$$\frac{\partial^2 x}{\partial\tau^2} + 2\epsilon\frac{\partial^2 x}{\partial\tau\partial\eta} + \epsilon^2\frac{\partial^2 x}{\partial\eta^2} + (\delta + \epsilon\cos\tau)x = 0 \tag{6}$$

(Feldman 2). We can expand x into a power series as follows:

$$x(\tau,\eta) = x_0(\tau,\eta) + \epsilon x_1(\tau,\eta)\ldots$$

Neglecting terms above ϵ^2, we can substitute x into (6), yielding:

$$\frac{\partial^2(x_0 + \epsilon x_1)}{\partial\tau^2} + 2\epsilon\frac{\partial^2(x_0 + \epsilon x_1)}{\partial\tau\partial\eta} + (\delta + \epsilon\cos\tau)(x_0 + \epsilon x_1) = 0.$$

We can now compare the constant terms and the ϵ terms, giving the two respective equations:

$$\frac{\partial^2 x_0}{\partial \tau^2} + \delta x_0 = 0, \tag{7}$$

$$\frac{\partial^2 x_1}{\partial \tau^2} + 2\frac{\partial^2 x_0}{\partial \tau \partial \eta} + \delta x_1 + x_0 \cos\tau = 0. \tag{8}$$

Notice that (7) is of the form of a simple harmonic oscillator, where $\sqrt{\delta}$ is the angular velocity. This is the benefit of perturbation, where the complex differential equation can be broken up into bits. The general solution to the equations can therefore be expressed as:

$$x_0 = A(\eta)\cos\sqrt{\delta}\tau + B(\eta)\sin\sqrt{\delta}\tau. \tag{9}$$

The amplitude of the pendulum oscillation is thus $\sqrt{A^2 + B^2}$. Notice how A, our amplitude is a function of η, while the inner cosine term is a function of t. Another thing we can see is this, by remembering our original definition for t:

$$\sqrt{\delta}\tau = \frac{\omega_p}{\omega_s}(\omega_s t) = \omega_p t.$$

This is something we can see experimentally later on. Substituting in x_0 into (8) and rearranging, we can get the following equation:

$$\frac{\partial^2 x_1}{\partial \tau^2} + \delta x_1 = -\frac{2\partial^2[A(\eta)\cos\sqrt{\delta}\tau + B(\eta)\sin\sqrt{\delta}\tau]}{\partial\tau\partial\eta} - \cos\tau(A(\eta)\cos\sqrt{\delta}\tau + B(\eta)\sin\sqrt{\delta}\tau).$$

We can then expand the right hand side by taking the partial derivative of τ and using the product to sum rule to get a general solution to our system:

$$\begin{aligned}\frac{\partial^2 x_1}{\partial \tau^2} + \delta x_1 =& 2\sqrt{\delta}\frac{dA}{d\eta}\sin\sqrt{\delta}\tau - 2\sqrt{\delta}\frac{dB}{d\eta}\cos\sqrt{\delta}\tau - A\cos\sqrt{\delta}\tau\cos\tau - B\sin\sqrt{\delta}\tau\cos\tau \\ =& 2\sqrt{\delta}\frac{dA}{d\eta}\sin\sqrt{\delta}\tau - 2\sqrt{\delta}\frac{dB}{d\eta}\cos\sqrt{\delta}\tau \\ &- \frac{A}{2}[\cos((\sqrt{\delta}+1)\tau) + \cos((\sqrt{\delta}-1)\tau)] \\ &- \frac{B}{2}[\sin((\sqrt{\delta}+1)\tau) + \sin((\sqrt{\delta}-1)\tau)].\end{aligned}$$

We can consider our particular case by the special value of $\delta = \frac{\omega_p^2}{\omega_s^2} = \frac{1}{4}$. The reason is because in order for the periods of the terms in the last two terms to be resonant, ie. have a angular velocity of $\sqrt{\delta}$, we need to solve the equation

$$|\sqrt{\delta} - 1| = \sqrt{\delta}.$$

Since δ is the square of the ratio between the two oscillators, $\frac{\omega_p^2}{\omega_s^2}$, this is equivalent to the spring having half the period of the pendulum, ie. $\omega_s = 2\omega_p$. Substituting δ, we get:

$$\frac{\partial^2 x_1}{\partial \tau^2} + \delta x_1 = \frac{dA}{d\eta}\sin\frac{\tau}{2} - \frac{dB}{d\eta}\cos\frac{\tau}{2} - \frac{A}{2}[\cos\frac{3\tau}{2} + \cos\frac{\tau}{2}] - \frac{B}{2}[\sin\frac{3\tau}{2} + \sin\frac{\tau}{2}].$$

We can then rearrange the right hand side into a more palatable form:

$$\frac{\partial^2 x_1}{\partial \tau^2} + \delta x_1 = -\frac{A}{2}\cos\frac{3\tau}{2} - \frac{B}{2}\sin\frac{3\tau}{2} + (\sin\frac{\tau}{2})(\frac{dA}{d\eta} + \frac{B}{2}) + (\cos\frac{\tau}{2})(-\frac{dB}{d\eta} - \frac{A}{2}).$$

By setting terms with frequency $\frac{\tau}{2}$ to zero,

$$0 = \sin\frac{\tau}{2}(\frac{dA}{d\eta} + \frac{B}{2})$$

$$0 = \cos\frac{\tau}{2}(-\frac{dB}{d\eta} - \frac{A}{2})$$

We can thus get:

$$\frac{dA}{d\eta} = -\frac{B}{2} \tag{10}$$

$$\frac{dB}{d\eta} = -\frac{A}{2} \tag{11}$$

It may also be observed that, for other values of δ, the derivatives of A and B would simply be equal to 0, resulting in a constant amplitude. This is because in that case, none of the terms proportional to A or B have a resonant frequency. By combining and substituting A and B in equations (10) and (11), we can say that

$$\frac{d^2 A}{d\eta^2} = \frac{A}{4}$$

$$\frac{d^2 B}{d\eta^2} = \frac{B}{4}.$$

The solutions are of the form

$$A(\eta) = \alpha e^{\frac{1}{2}\eta} + \beta e^{-\frac{1}{2}\eta}$$

$$B(\eta) = \alpha e^{\frac{1}{2}\eta} + \beta e^{-\frac{1}{2}\eta}$$

for α, $\beta \in R^+$. We see that A and B must grow exponentially. Therefore, given that the ratio of the two oscillators is 1:2, the horizontal amplitude $\sqrt{A^2 + B^2}$ of the pendulum oscillation must also grow exponentially as the positive term quickly dominates when $n \neq 0$.

Obviously, the pendulum cannot simply grow exponentially when left on its own. The length of the pendulum changes as its amplitude changes. This is because as the amplitude and hence the angle to the vertical of the pendulum rod increases, so too does the tension along the rod. This results in the spring being stretched more than it was originally, extending the length and thus changing the ratio between the two angular velocities. However, its amplitude does indeed grow exponentially when the rod is externally excited periodically, for example by tugging the string and thus adding energy and driving to the system.

Now, we shall consider a small desynchronization in δ, away from $\frac{1}{4}$, by expanding in a power series of ϵ:

$$\delta = \frac{1}{4} + \delta_1\epsilon + \delta_2\epsilon^2 + \dots \tag{12}$$

This expression for δ then is substituted into equation (8):

$$\frac{\partial^2 x_1}{\partial \tau^2} + \frac{1}{4}x_1 = -2\frac{\partial^2 x_0}{\partial \tau \partial \eta} - x_0 \cos\tau - \delta_1 x_0 \tag{13}$$

We follow a similar process as before, substituting the general solution for x_0 stated in (9) into the right hand side of (13) to get:

$$\begin{aligned}\frac{\partial^2 x_1}{\partial \tau^2} + \frac{1}{4}x_1 &= 2(\frac{1}{2}\frac{dA}{d\eta}\sin\frac{\tau}{2} - \frac{1}{2}\frac{dB}{d\eta}\cos\frac{\tau}{2}) \\ &\quad - A\cos\frac{\tau}{2}\cos c - B\sin\frac{\tau}{2}\cos\tau - A\delta_1\cos\frac{\tau}{2} - B\delta_1\sin\frac{\tau}{2} \\ &= \frac{dA}{d\eta}\sin\frac{\tau}{2} - \frac{dB}{d\eta}\cos\frac{\tau}{2} - \frac{A}{2}(\cos\frac{3\tau}{2} + \cos\frac{\tau}{2}) - \frac{B}{2}(\sin\frac{3\tau}{2} - \sin\frac{\tau}{2}) \\ &\quad - \delta_1 A\cos\frac{\tau}{2} - \delta_1 B\sin\frac{\tau}{2}\end{aligned} \tag{14}$$

Because we need the resonance terms - $\sin\frac{\tau}{2}$ and $\cos\frac{\tau}{2}$ - to be zero, we ignore the other terms to get:

$$0 = \sin\frac{\tau}{2}(\frac{dA}{d\eta} + \frac{B}{2} - \delta_1 B)$$

$$0 = \cos\frac{\tau}{2}(\frac{dB}{d\eta} + \frac{A}{2} - \delta_1 A)$$

Similar to (10) and (11):

$$\frac{dA}{d\eta} = (\delta_1 - \frac{1}{2})B \tag{15}$$

$$\frac{dB}{d\eta} = -(\delta_1 + \frac{1}{2})A \tag{16}$$

Then taking the derivative of (15) with respect to η and substituting (16):

$$\frac{d^2 A}{d\eta^2} = (\delta_1^2 - \frac{1}{4})A$$

Now, we have two cases:

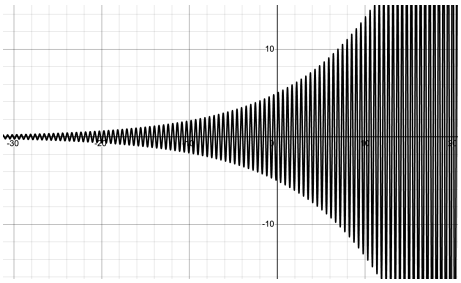
1. $\delta_1 > \frac{1}{2}$ or $\delta_1 < -\frac{1}{2}$, which means $\frac{d^2A}{d\eta^2} > 0$
2. $-\frac{1}{2} < \delta_1 < \frac{1}{2}$, which means $\frac{d^2A}{d\eta^2} < 0$

The first case implies that the amplitude is an exponential function, along the same line of logic as the previous example. The second case implies a sinusoidal change of amplitude, as the second derivative is negatively proportional to itself. Ignoring any other factors, such as the change in length, Case 1 would be in the form of Figure4a, and Case 2 would be in the form of Figure4b.

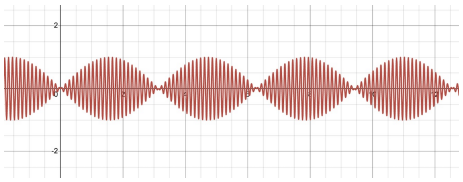
Returning to (12), we plot this boundary in Figure 5 on the parameter space of δ and ϵ:

$$\delta = \frac{1}{4} \pm \frac{\epsilon}{2} + O(\epsilon^2). \tag{17}$$

We see that for epsilon–delta combinations above the boundary, the predicted behaviour will be of Case 1, i.e. "unstable". Below will be of Case 2, stable.



(a) An arbitrary example of exponential growth, $y = 5e^x \sin 12x$.



(b) An arbitrary example of sinusoidal growth, $y = \sin 50x \sin x$

Figure 4: Exponential vs SHM growth

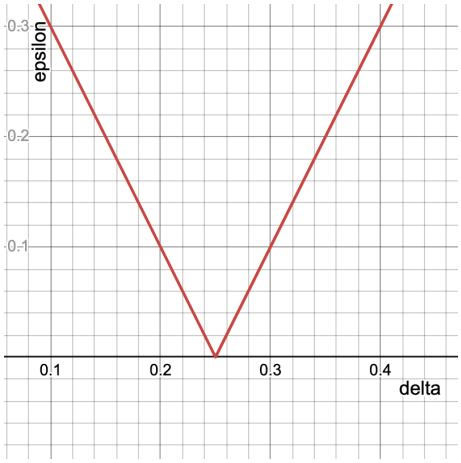


Figure 5: Plot of equation (17). Values of δ and ϵ enclosed between the two lines

2.2 Result

The significance of Figure 5 is to show that with a constant ϵ, as δ varies, the point will slide between the stable and unstable regions. This reflects the observed phenomenon, where $\delta = \frac{\omega_p^2}{\omega_s^2}$ changes in value as the amplitude of the horizontal oscillation changes. This is because while z_0 is the average value of l when the pendulum amplitude is small, the average value is longer when the pendulum amplitude is large, resulting in a change in δ. This change is observed in our experiment.

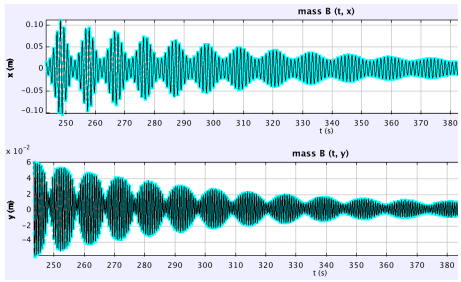


Figure 6: Horizontal oscillation (top) and vertical oscillation (bottom).

An illustrative example of this behaviour is in the comparison between the horizontal and vertical oscillations, seen in Figure 6, taken from our experiment. While the horizontal oscillations, as we know, are unstable, by observation we can see that the vertical oscillations follow our pre-

dictions for an oscillation in the stable region. While the horizontal oscillations grow and decay exponentially, resulting in "sharp" peaks, the vertical oscillations grow and decay sinusoidally, resulting in "round" peaks. This is in accordance with our findings illustrated in Figure 4 and 5, describing the behaviour of values of δ within and outside of the area of stability. This is the root cause of the changing amplitude of the pendulum, provided initially $\delta = \frac{1}{4}$.

In summation, by approximating the vertical spring motion, we derived an equation for the motion of the mass in the x-direction. This equation is a differential equation known as Mathieu's equation, and by use of perturbation theory, we find that $\delta = \frac{1}{4}$ results in resonance, from which we can find the form of our expressions for the amplitude, $A(\eta)$ and $B(\eta)$. We will now attempt to confirm this value of δ experimentally.

2.3 Range of Frequencies

We will explore the range of frequencies outside of, but near $\delta = \frac{1}{4}$ through the use of eigenvalues. We can first introduce the angular velocity ω_n, which is the natural or ideal frequency for the pendulum. This value is close to, but not necessarily equal to the pendulum angular velocity ω_p. We start with the transformed Mathieu's equation

$$\frac{d^2x}{dt^2} + \omega_n^2[1 + f(t)]x = 0,$$

where the transformed variable $x(t) = \alpha(t)\cos(\omega_p t) + \beta(t)\sin(\omega_p t)$, $f(t) = a\sin(2\omega_p t)$. We can substitute these to get the following system of linear differential equations, then use eigenvalue/eigenvector methods to decouple them:

$$2\omega_p \frac{d\alpha}{dt} = \frac{1}{2}a\omega_n^2\alpha - (\omega_p^2 - \omega_n^2)\beta,$$

$$2\omega_p \frac{d\beta}{dt} = (\omega_p^2 - \omega_n^2)\alpha - \frac{1}{2}a\omega_n^2\beta.$$

These can be written in the general form of $\dot{x} = Ax$, that is

$$\frac{d}{dt}\begin{bmatrix}\alpha\\ \beta\end{bmatrix} = \frac{1}{2\omega_p}\begin{bmatrix}\frac{1}{2}a\omega_n^2 & -(\omega_p^2 - \omega_n^2)\\ (\omega_p^2 - \omega_n^2) & -\frac{1}{2}a\omega_n^2\end{bmatrix}\begin{bmatrix}\alpha\\ \beta\end{bmatrix}.$$

The eigenvalue equation for pair (x, λ) is $Ax = \lambda x$, or $(A - \lambda I)x = 0$. Thus we need to find $A - \lambda I = 0$, where I is the identity matrix. In our case this is

$$A - \lambda I = \begin{bmatrix}\frac{1}{2\omega_p}(\frac{1}{2}a\omega_n^2) - \lambda & -\frac{1}{2\omega_p}(\omega_p^2 - \omega_n^2)\\ \frac{1}{2\omega_p}(\omega_p^2 - \omega_n^2) & -\frac{1}{2\omega_p}(\frac{1}{2}a\omega_n^2) - \lambda\end{bmatrix}.$$

The determinant of this is

$$\begin{aligned}\det(A - \lambda I) &= [(\frac{1}{2\omega_p}(\frac{1}{2}a\omega_n^2) - \lambda)(-\frac{1}{2\omega_p}(\frac{1}{2}a\omega_n^2) - \lambda)] + (\frac{1}{2\omega_p})^2(\omega_p^2 - \omega_n^2)^2\\ &= (\frac{1}{2\omega_p})^2[-(\frac{1}{2}a\omega_n^2)^2 + \lambda^2(2\omega_p)^2 + (\omega_p^2 - \omega_n^2)^2].\end{aligned}$$

Setting the determinant to 0 as desired, we can isolate the eigenvalue λ to get

$$\begin{aligned}\lambda^2 &= (\frac{1}{2\omega_p})^2[(\frac{1}{2}a\omega_n^2)^2 - (\omega_p^2 - \omega_n^2)^2]\\ \lambda_{1,2} &= \pm\frac{1}{2\omega_p}\sqrt{(\frac{1}{2}a\omega_n^2)^2 - (\omega_p^2 - \omega_n^2)^2}.\end{aligned}$$

If we let $\Delta\omega = \omega_p^2 - \omega_n^2$ and rewrite ω_p as ω_n, we obtain

$$\lambda_{1,2} = \pm\sqrt{(\frac{a\omega_n}{4})^2 - \Delta\omega^2}.$$

As such, if $|\Delta\omega| < \frac{a\omega_s}{4}$, there is one positive, real eigenvalue, leading to exponential growth in $\alpha(t)$ and $\beta(t)$, which is the condition for parametric resonance. This range is significant - ω_p does not necessarily have to equal ω_n for the resonance to occur; but the inequality has to be satisfied.

2.4 Literature Review

2.4.1 Existing Work

A key contribution to the Lagrangian method used above is Olsson's *Why does a mass on a spring sometimes misbehave?*. It outlines the approach to the derivation, specifically setting up the Lagrangian, and taking the 'dominant spring mode', ie. $x \ll z$ [5].

Another important reference we used is *Mathieu's Equation and Its Generalizations: Overview of Stability Charts and Their Features* by Kovacic et al, a detailed analysis of different approaches to Mathieu's equation in particular. This was helpful in reaching the critical result in (17) [4].

There is existing research on this topic, such as Vitt and Gorelik's *Oscillations of an Elastic Pendulum as an Example of the Oscillations of Two Parametrically Coupled Linear Systems*, which concentrates on the 2:1 ratio, finding two solutions using secular perturbations, and makes note of energy transfer between the pendulum and spring modes [8]. However, this source, like many more, are very mathematical with little to basic experimental verification. Others, such as Giblin's *Stepwise Precession of the Stochastic Swinging Spring* [1] or Holm's *Stepwise Precession of the Resonant Swinging Spring* [2], use simulations rather than empirical methods. Thus we wish to combine existing work on the Lagrangian and Mathieu equations to understand the resonant behaviour of the elastic pendulum, and collect experimental data to empirically verify our theory.

2.4.2 Precession

We found Holm (2001) to be the main contributor to the investigation of the precession observed in the elastic pendulum. While the motion of the mass during its horizontal swinging phase is generally planar, we have observed that the plane in which each horizontal oscillation occurs changes as it decays and reappears. They claim that this precession occurs during the vertical oscillation phase of the mass's motion, and it is predictable via a formula they derive.

The modulation equations they derive from the approximate Lagrangian resemble the three-wave equation, which describe a wide range of small-amplitude resonant waves. This reflects the capability for the elastic pendulum to simulate different real-life phenomena, such as behaviour of plasma or light waves in a heterogeneous medium. A model was thus derived using coordinates which rotate at specific critical angular velocities.

3 Experiment

3.1 Experimental Setup and Equipment

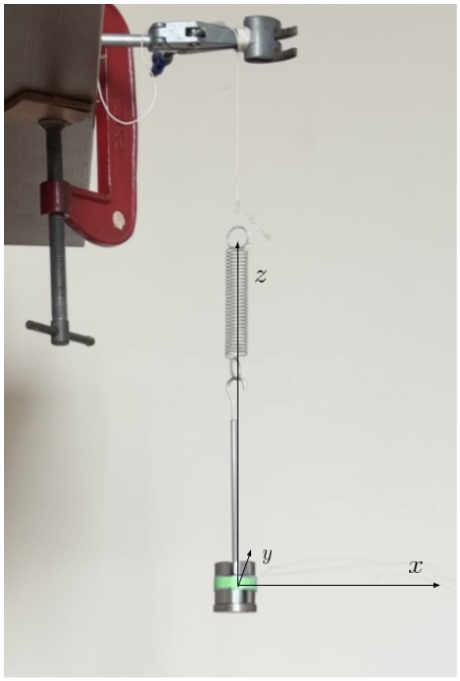


Figure 7: Experiment setup with relevant coordinate axes

Our setup can be seen in Figure 7. A weight of 200 grams is tied to a spring of spring constant 30 N/m and a piece of string, forming the pendulum rod. By varying the length of the string, the rest length of the pendulum rod z_0 may be varied.

Three cameras are mounted perpendicular to each other, centered by the initial position of the mass. After calibrating the length, a 2-minute video is then taken of the motion of the mass, and then analysed using Tracker. The green strip is stuck onto the center of mass as the point to track.

It is worth noting that the weight is not a point mass, and has a length that must be considered. Thus the data of l_0 and z_0 have all been adjusted to be the length from the top of the string to the center of mass, instead of the base of the weight.
There are a few important things when selecting equipment:

- A shorter weight is highly preferred: the weight shown in Figure 7 has a very long hook, thus limiting the lower bound of our independent variable. We exchanged this for a shorter weight of equal mass.
- In theory, we referred to the pendulum as a 'rod', as this would be ideal. With the string that we used, we noticed a slight imperfection as it is slightly elastic. We hence measured experimental values for the period of the spring to measure its deviation from the theoretical value k/m, which was negligible. Additionally, we noticed the weight occasionally spinning, which could have affected its motion. However, these pitfalls are mitigated by the fact that the string is far easier to adjust in length and can be minimised through careful experimentation.

3.2 Variables

Our independent variable is the length of the string - along with the spring, they form the pendulum rod. The resonant behaviour is dependent on the ratio between the angular velocities of the oscillators, $\delta = \omega_p^2/\omega_s^2$. Since $\omega_s^2 = k/m$ is fixed, we adjust $\omega_p^2 = g/z_0$ to study the system's behaviour at different values of δ. The experimental values for z_0 range from 16.8 to 55.4 cm.

Our dependent variable is the amplitude of the pendulum, as measured by its angle to the vertical. Since our initial displacement is nearly vertical, this is effectively a measurement for how much of the spring energy has been transferred to the pendulum system. We can calculate the angular frequencies of the system by calculating the periods of the oscillations. We can thus measure the optimal ratio, such that the energy is maximally transferred between the two systems. One flaw is parallax error, where the amplitude of long pendulums is exaggerated. However, this is reduced by placing the camera as far away from the pendulum as possible.

3.3 Methodology

1. As shown in Figure 7, securely clamp a retort clamp to a desk, ensuring there is ample room for the mass to swing underneath.
2. Orient three cameras perpendicular to each other - one directly below, and two level with the resting mass.
3. Mark the string at a length of 3cm, tying it on the retort clamp.
4. Attach the weight to the string and begin recording on all cameras.
5. Hold a meter stick to the side of the weight with respect to each camera. This is to serve as a reference for the tracking software.
6. Lift the weight as vertically as possible, to the point where the spring is at its rest length without the mass (to ensure equal initial potential energy). With a constant mass, this ensures the same initial displacement for each trial. This is preferable to horizontal displacement due to ease of measurement, accuracy, and as the pendulum is very sensitive to initial horizontal displacements.
7. Release the weight
8. 2 minutes later, repeat steps 6-7 twice.
9. Repeat the process for strings of various lengths up to 29 cm[1].

[1]The specific lengths were not uniform, rather selected to paint a sufficient picture - we chose large intervals with longer lengths far from the 2:1 ratio, while we took many trials at small intervals closer to the optimal length.

4 Experimental Data

4.1 Extracting the Period

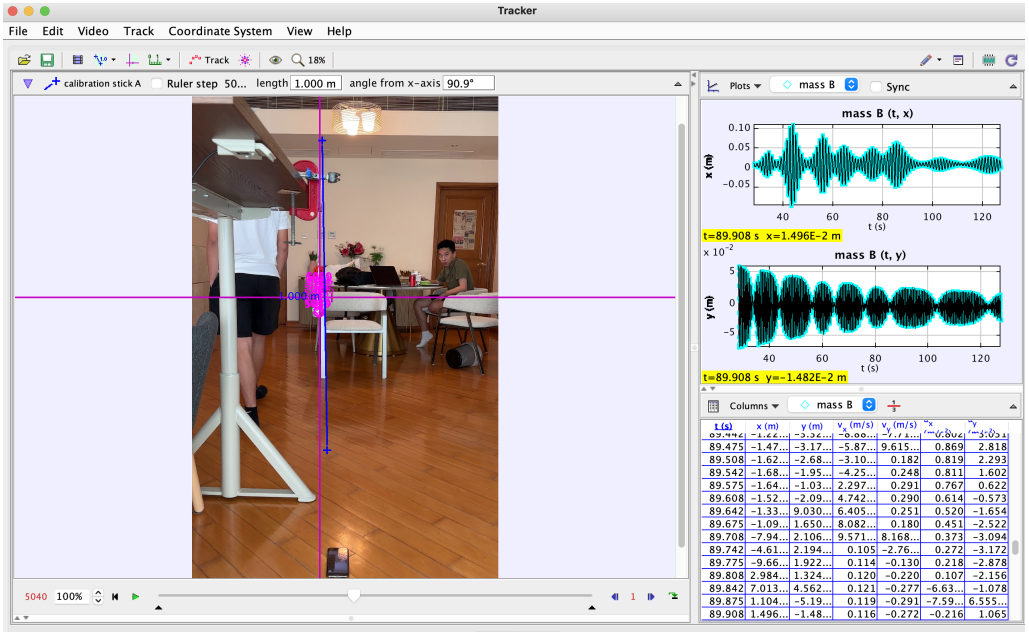


Figure 8: Example of Tracker application interface. The meter calibration is shown in blue, and each tracked data point is marked in pink, with the coordinate axes intersecting at its initial position. This data is then collected in the table and graph on the right. Notice the shape of the graph. The erratic amplitudes of the horizontal oscillation is due to precession.

After gathering all the experimental footage, we upload it all to Tracker [7] to track the mass's position frame by frame, shown in Figure 8. The mass's horizontal and vertical position is plotted against time. One point of note is that in Tracker, vertical positions were saved under the variable y. Using the Autotrack feature, we can export the data to arrange in a spreadsheet. To extract the vertical and horizontal periods, we use the conditional formatting feature, applying a custom formula[2] to the columns of all x and y. This will highlight the cell with the minimal value of each interval of some set number of cells, shown in Figure 9. This is equivalent to measuring the time between each oscillation by marking each time the pendulum reaches its leftmost position, or equivalently minimum value for the displacement. We changed this number manually based on the apparent period of the data, then went through manually to correct it.

Notice that there are clear errors in these highlighted cells, so after filling/unfilling erroneous cells manually, it should look something like Figure 10. Noticeably, highlighted cells appear periodically within each column. With this, we can use the Apps Script extension to count the number of cells between subsequent highlighted cells. The code is shown in Figure 17 in the appendix.

This code will output a list of numbers, which we can put into a document, shown in Figure 11. This also acts as a check for errors, manually evaluated and corrected. The list is then processed by taking the mean, highlighted in orange.

[2]The custom formula used is **=AND($DJ1=MIN(OFFSET($DJ$1, FLOOR((ROW($DJ1)-1)/33, 1)*33, 0, 33)), ROW($DJ1)<=COUNT($DJ:$DJ)-33)**. This is for column DJ and highlighting the minimum for each interval of 33 rows. These values can be changed for each set of data.

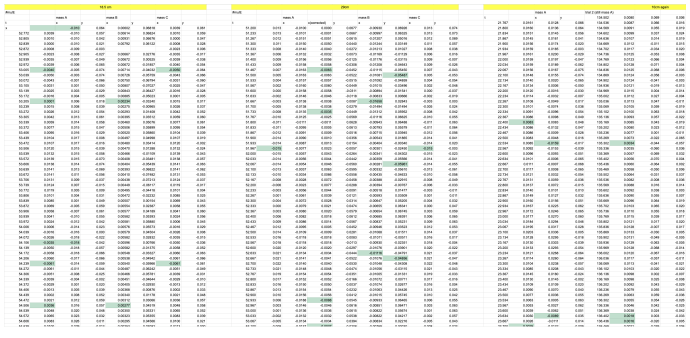


Figure 9: Spreadsheet after conditional formatting

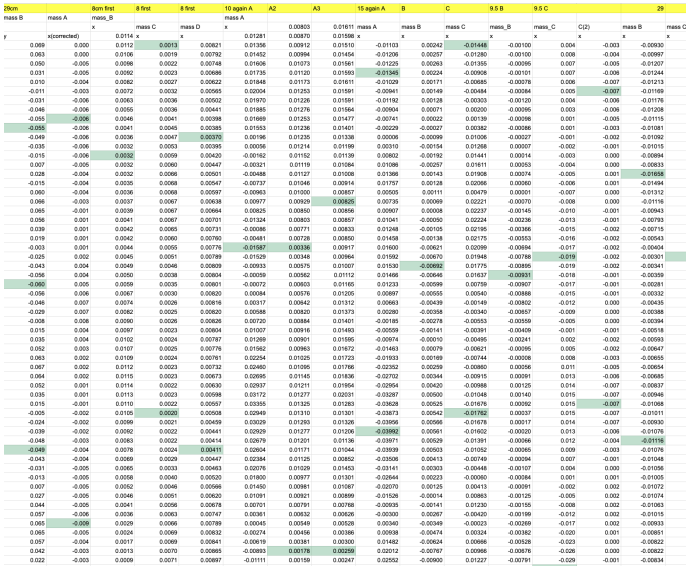


Figure 10: Processed spreadsheet

4 cm

Mas A x:
9,
33,**33**,32,**33**,**33**,**34**,40,35,**34**,**33**,**33**,**34**,**33**,**34**,35,38,35,32,32,32,32,**34**,**33**,**33**,**34**,**33**,**34**,**33**,**34**,35,37,**34**,
33,**33**,**33**,31,32,32,**33**,**34**,**33**,**33**,**33**,**33**,**33**,**33**,31,32,**33**,32,32,**34**,36,35,35,**33**,**34**,**33**,**33**,32,32,**33**,
32,32,**33**,**33**,32,32,36

Mean (Average)33.369230769231
Median 33
Range 9
Mode 33, appeared 26 times
Min 31

10 cm

Mass A x 9,
32,36,**33**,32,**33**,**34**,**33**,35,39,**33**,40,35,**34**,**33**,**33**,**34**,**34**,**34**,31,**34**,**34**,30,35,**34**,**34**,35,43,35,35,
34,**33**,35,**33**,**34**,39,38,36,35,**34**,**34**,**34**,**33**,36,36,38,37,35,**34**,**34**,**34**,**34**,35

Mean (Average)34.653846153846
Median 34
Range 13
Mode 34, appeared 18 times
Count: 18/52 =0.346

10 again

Mass A(1) x: 21,
35,35,**33**,**34**,**33**,**34**,35,43,35,35,35,**33**,**34**,**34**,**34**,35,40,38,35,**34**,**34**,**34**,**33**,35,**33**,**34**,32,32,36,
34,36,39,36,35,**34**,**34**,**34**,35,**34**,**34**,37,39,36,35,35,**34**,35,35,35,**34**,**33**,36,37,36,41,32,30,36,35,
31,35,35,**34**,**34**,**34**,35,35,35,38,37,35,35,35,**34**,35,**34**,35,35,36,37,37,35,35,35,35,35
,34

Mean (Average)34.988235294118
Median 35

Figure 11: Google doc with the outputs from the code

Finally, we can take the mean period of each trial of each length, leaving us with results shown in Table 1:

l_0 (m)	Mean Period (Frames, 1/30 s)	Period (s)
0.101	24.853	0.828
0.132	26.881	0.896
0.167	28.154	0.938
0.179	30.160	1.005
0.193	33.164	1.105
0.209	33.321	1.111
0.227	33.525	1.103
0.237	34.061	1.146
0.277	36.128	1.185
0.287	36.626	1.204
0.292	36.873	1.203
0.297	37.118	1.196
0.312	37.415	1.247
0.317	39.484	1.230
0.347	39.484	1.251
0.382	41.060	1.321
0.487	45.460	1.481

Table 1: Measured pendulum periods, for different values of l_0

We now want to graph the unstretched length l_0 against the pendulum period. Using $\omega_p = \sqrt{\frac{g}{z_0}}$, we can get

$$\frac{2\pi}{T_p} = \sqrt{\frac{g}{l_0 + \frac{mg}{k}}}.$$

Plugging in our values, this is

$$\frac{2\pi}{T_p} = \sqrt{\frac{9.8}{l_0 + \frac{0.2 \cdot 9.8}{30}}}.$$

Indeed, our data in Figure 12 fits this well, as we expect, since our solution for x has a frequency of $\sqrt{\delta}\tau = \omega_p t$. We can use this line to interpolate between data points. The error bars chosen are ± 0.5 cm horizontally, and $\pm \frac{2}{30}$ s vertically.

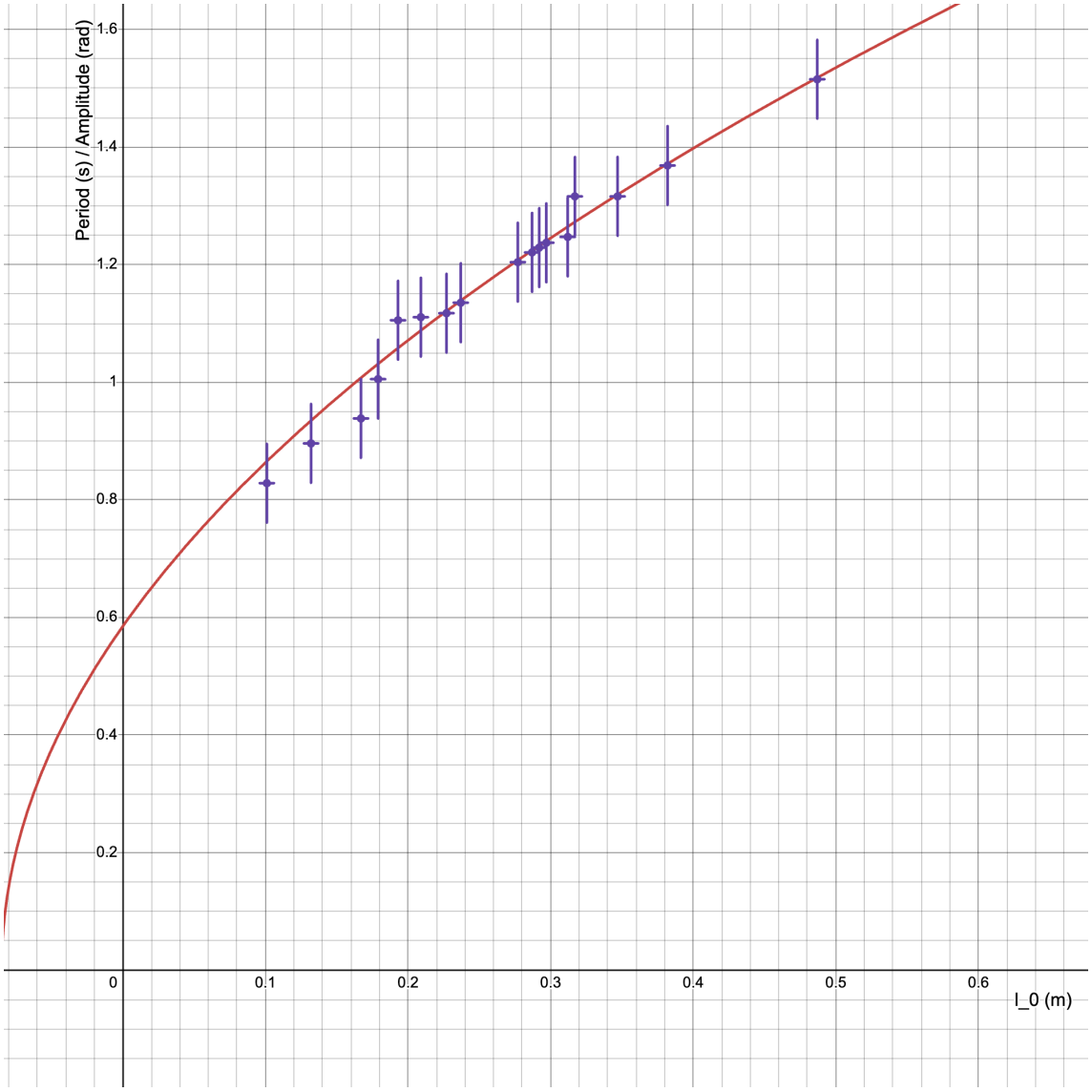


Figure 12: Graph of l_0 and horizontal period

4.2 Extracting the pendulum amplitude

The next part of the data processing involves the amplitude of the oscillation. This is much more straightforward, finding the maximum displacement on the xy plane (parallel to the floor) by the bottom camera using the Pythagorean theorem. With the displacement, we will then use it to find the angular amplitude, given by $\arcsin(\frac{\text{max displacement}}{z_0})$, shown in Table 2:

l_0 (m)	Average Maximum Displacement (m)	Average Angle Amplitude
0.101	0.0163	0.0972
0.109	0.0115	0.0651
0.132	0.0145	0.0727
0.167	0.0116	0.0496
0.179	0.0162	0.0660
0.193	0.0274	0.1056
0.209	0.0694	0.2543
0.227	0.0885	0.3057
0.235	0.0924	0.3108
0.237	0.0905	0.3023
0.258	0.1093	0.3431
0.277	0.1101	0.3258
0.287	0.1196	0.3445
0.292	0.1186	0.3368
0.297	0.1273	0.3572
0.312	0.1197	0.3214
0.317	0.1040	0.2742
0.347	0.0859	0.2089
0.382	0.0377	0.0840
0.487	0.0290	0.0523

Table 2: Measured amplitudes for different values of l_0

The optimal lengths will then correspond to the point where $\omega_s = 2\omega_p$, or more conveniently, $T_p = 2T_s$. In Figure 13, these are shown as lines, respectively $T_p = 2(\frac{17.6}{30})$ (green) and $T_p = (\frac{17.6}{30})$ (orange), where $T_s = \frac{17.6}{30}$ is found experimentally as the number of frames on 30 fps. We are focusing on the $\omega_s = 2\omega_p$ case, which is when

$$\frac{2\pi}{2 \cdot \left(\frac{17.6}{30}\right)} = \sqrt{\frac{9.8}{x + \frac{0.2 \cdot 9.8}{30}}}$$

or $l_0 \approx 0.2567$ (purple).

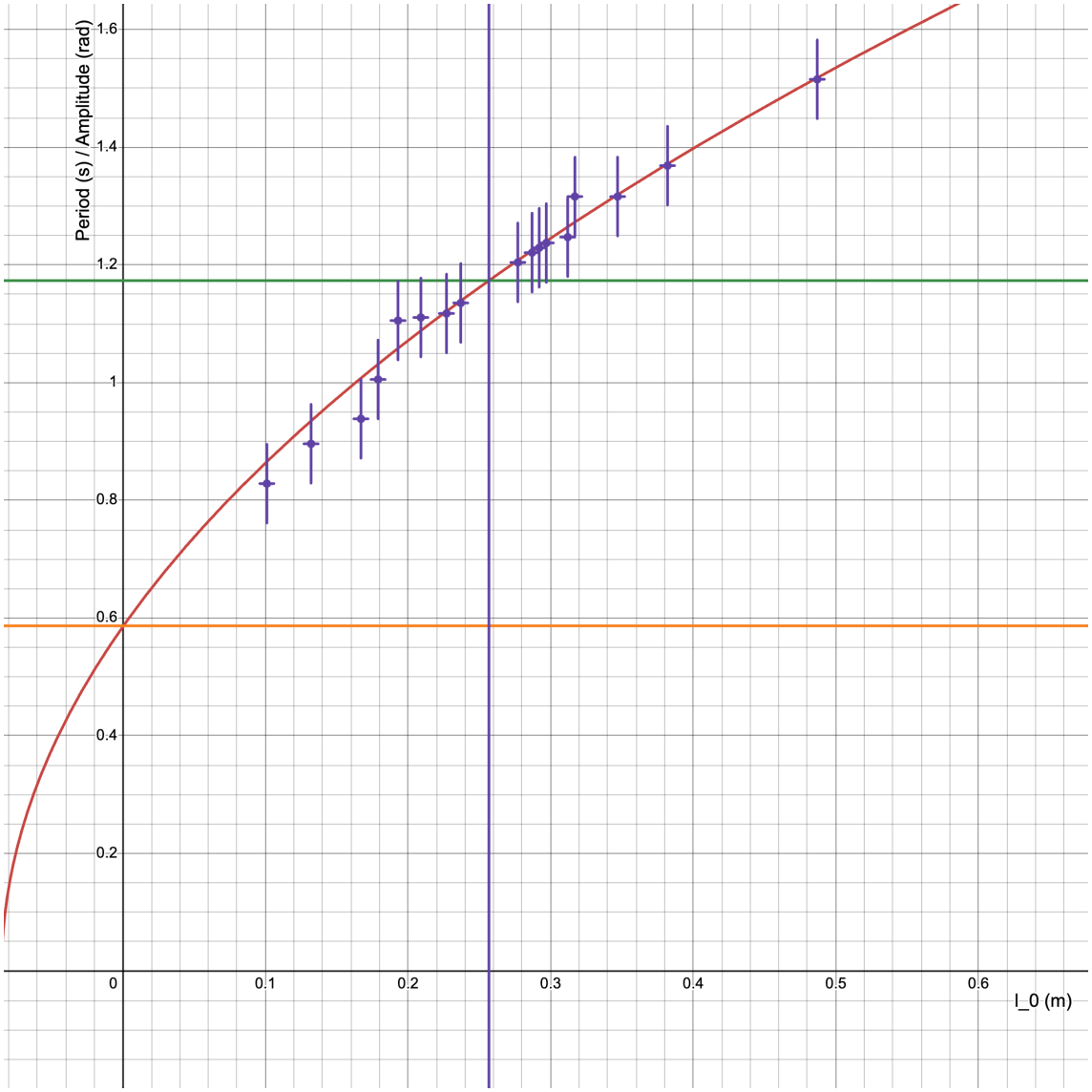


Figure 13: Table 1 graphed. Three lines are now overlaid on the original graph. The intersection between the green and red lines represents our critical value for l_0, 25.7 cm and the corresponding pendulum period, 1.173 seconds.
- Period where $\omega_s = \omega_p$ is marked in orange.
- Period where $\omega_s = 2\omega_p$ is marked in green.
- Line of best fit marked in red.
- Intersection between line of best fit and the optimal period is marked in purple.

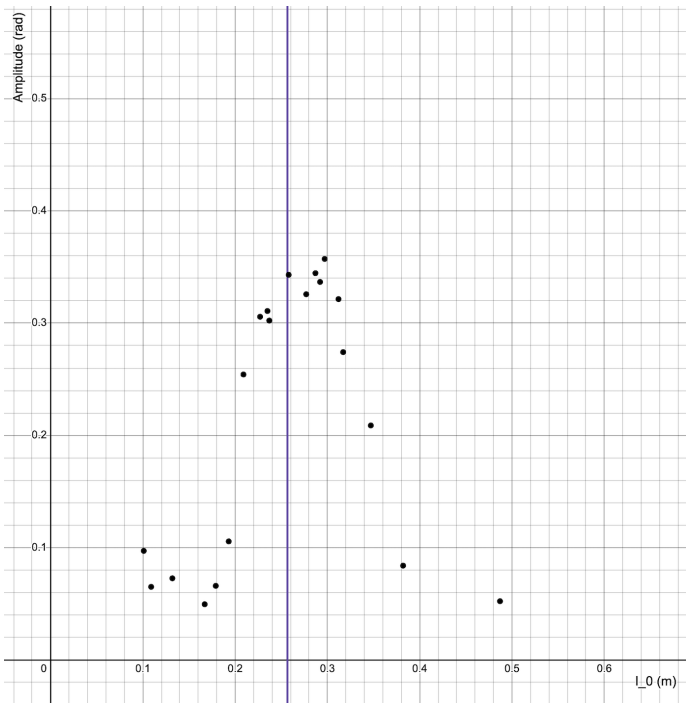


Figure 14: Table 2 graphed. The vertical purple line is the same as that in Figure 13, referring to the optimal length l_0, 25.7 cm.

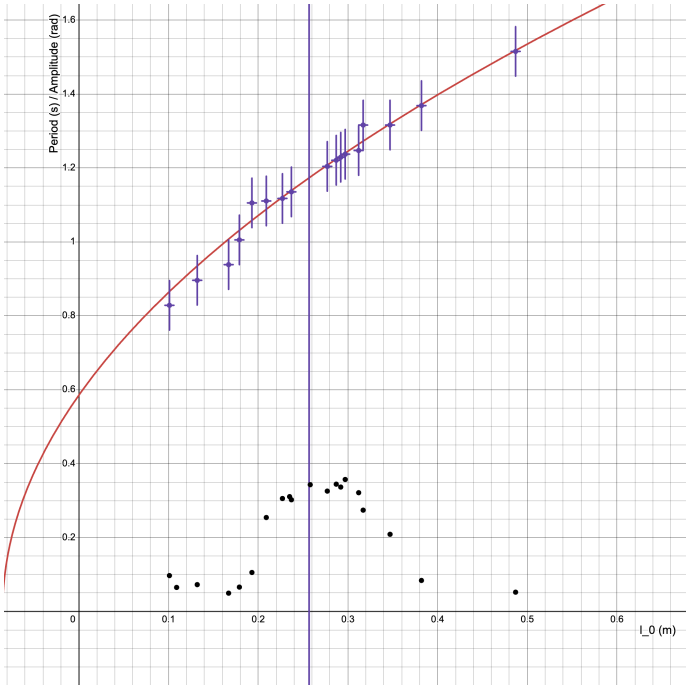


Figure 15: Amplitude and period graphed together, notice the amplitude's peak lines up with the optimal length given by the purple line.

4.3 Findings from graphs

Figure 13 represents the way in which we find the corresponding value of l_0, the unstretched length of the rod, where the ratio $\omega_s = 2\omega_p$ is satisfied. We collected period data using Tracker as described above, by calculating the average amount of frames between every minimum in the x-position. By plotting periods against l_0, we verified the square root relation that we expect from our derivation, as $\omega_p^2 = g/z_0$ where $z_0 = l_0 + \frac{mg}{k}$, and m, g and k are constant.

We then used our other relation, $\omega_s^2 = k/m$ to derive the pendulum period for which $\omega_s = \omega_p$. This value is at 0.5865 seconds. By doubling this value, we can find the pendulum period where $\omega_s = 2\omega_p$, 1.173 seconds. The point at which this line, marked in green, intersects with our line relating period and l_0, marked in red, represents the optimal value for l_0. This value is calculated to be at 25.7 cm.

Figure 14 plots the amplitudes in radians against l_0. Since we know that theoretically, 25.7 cm represents the optimal value of l_0, this line is plotted on the graph, marked in purple. We can thus see that amplitudes exhibits a maximum at a value of l_0 equal to our theoretical value. It is also noteworthy that there are multiple high amplitude values on either side of l_0, as expected.

4.4 Notes on Precession

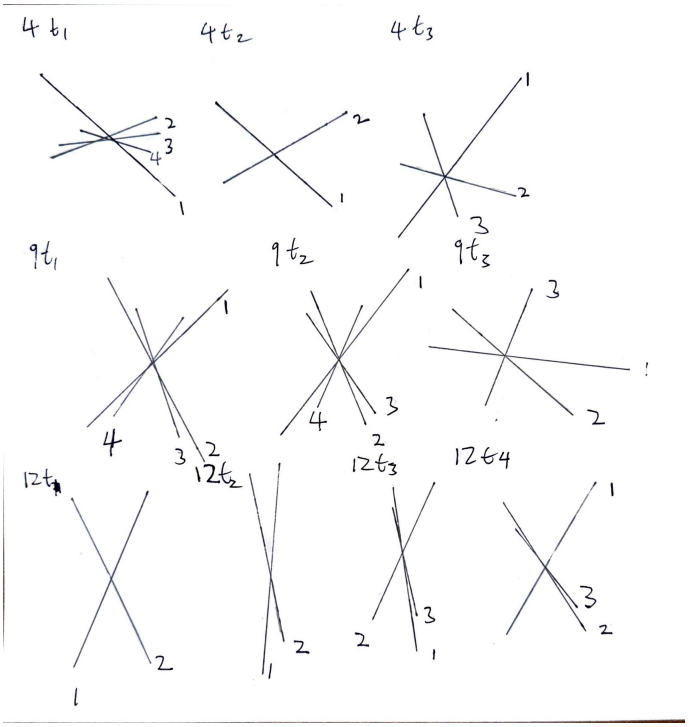


Figure 16: Several planes of oscillation - trials taken at string lengths of 4, 9 and 12. Handdrawn for qualitative assessment.

Figure 16 illustrates some planes of oscillation that we traced for strings of length 4, 9, and 12cm (at and around our optimal lengths). It sometimes seems regular across trials, relatively consistent for the first and second planes. However, as shown, the remaining precessions can be quite haphazard, and we can't draw a significant conclusion in agreement with the literature

discussed above. In addition, for string lengths farther from our optimal length for the 2:1 ratio, the oscillations become more circular, and the spring/pendulum modes are not distinct, which does not allow for clear planes to compare. This corroborates the simulated results from Giblin [1] and Holm [2], where the mass travels in an ellipse, whose eccentricity varies based on the behaviour of the pendulum.

5 Conclusion

Our experiment has verified the theory as spelled out in the derivation. To summarise what we expect:

1. By analysing resonance in Mathieu's equation, we have found that our corresponding value for $\sqrt{\delta}$ should be $\omega_p/\omega_s = 1/2$ in order to cause resonant behaviour.
2. We expect the greatest amplitude growth at the corresponding length where this ratio is satisfied, where $l_0 = 25.7$ cm, and our oscillation has an average period of 1.17 seconds
3. According to the eigenvalue derivation however, we also expect this resonant behaviour to exist not only at the critical frequency ratio, but also nearby. It will thus span a range of pendulum lengths on either side of our derived value l_0.

Experimentally, we found the length that corresponds to the maximum amplitude in Figure 15, which aligns with the length where $2\omega_p = \omega_s$, shown in purple. This length is also at the centre of the range of values for which resonance is observed. The range is expected as derived through the eigenvalue method, where ω_p simply has to be within a range away from ω_n. The length of the pendulum is related to the period by a square root relation, as defined in our theory: $\omega_p = \sqrt{\frac{g}{z_0}}$. This relation is further verified by our experimental data regarding the periods of each pendulum oscillation. As seen in Figure 11, we can see the period of the pendulum changing significantly in regular intervals when transitioning between vertical and horizontal oscillation modes, where the bolded periods highlight when $\omega_s \approx 2\omega_p$. This reflects the ability for the system parameters, ie. the length of the pendulum, to oscillate around the resonant value.

About uncertainty, while each trial is accurately tracked by a computer, there is some deviation between trials. As a whole, uncertainty is difficult to measure and control throughout the experiment. This is because it is simply impossible to start at precisely the same location, and this system amplifies and hence is extremely sensitive to initial horizontal displacement. In fact, a distinctive feature of parametric resonance as opposed to normal resonance is that given that the starting value for x and $\dot{x}$ is 0 ie. a perfect launch, it genuinely will not deviate horizontally. However, this value is unstable and in practice, any deviation from the vertical axis will lead to growth in amplitude. Despite this, the general trend can be discerned through averaging multiple trials, and behaviour is remarkably consistent - while the precise motion is not predictable, aspects such as period and amplitude are generally consistent between trials. To improve experimental precision, an automated release mechanism could be used to launch the mass in future studies.

Since our experiment captured the motion in 3D, our results can be used to analyse additional aspects of the dynamics beyond planar resonance. The precession described earlier was observed, but not at a sufficiently consistent or prominent manner. In our experiment, we observed some trials having no precession at all, some precessing between two perpendicular planes, and others precessing erratically. This could be due to an asymmetrical setup on our part, such as in the way the mass hooks onto the spring, or the twist in the string, causing specific directions to be favored. This can be an area of further investigation, along with numerical analysis via simulations. Another area of future study could be the chaotic motion of the pendulum, and its sensitivity to initial conditions, which we have experimentally observed but not quantified. Its correspondence

to other dynamical systems in other contexts has also been noted by some sources [8] [2] and thus can be a worthwhile area of study, utilising the same mathematical tools.

In conclusion, by using perturbation theory in conjunction with the approximate Lagrangian of the elastic pendulum, as well as analysing its regions of stability, we quantitatively derived the main resonant angular velocity combination to achieve parametric resonance: $\omega_s = 2\omega_p$. Since ω_p is related to the length, and the length is related to the amplitude, the amplitude of the horizontal oscillations is itself oscillating. This is intuitively understandable as it reflects the spring extending every time the pendulum is at maximum velocity. Using Tracker, we measured the maximum amplitude of each pendulum oscillation and plotted it against the length of the pendulum rod. We have thus found the length at which energy is transferred maximally between the two modes of oscillation, and verified the theoretical relation.

References

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A Appendix

```
1  function countHighlightedRows() {
2    var sheet = SpreadsheetApp.getActiveSpreadsheet().getActiveSheet();
3    var dataRange = sheet.getDataRange();
4    var values = dataRange.getValues();
5    var numRows = values.length;
6    var numCols = values[0].length;
7    var highlightColor = "#a4c2f4"; // Color code of your highlighted cells
8
9    var count = 0;
10   var highlightedRows = [];
11
12   for (var row = 5; row < numRows; row++) {
13     var cell = sheet.getRange(row + 1, 5); // Change the column number as needed
14
15     // Check if the cell background color matches the highlight color
16     if (cell.getBackground() === highlightColor) {
17       if (count > 0) {
18         highlightedRows.push(count);
19         count = 0;
20       }
21     } else {
22       count++;
23     }
24   }
25
26   Logger.log("Highlighted rows count: " + highlightedRows.length);
27   Logger.log("Counts: " + highlightedRows);
28 }
```

Figure 17: Code to count number of rows between highlighted cells, changing the column number (line 13) and color code (line 7) as needed.